

Experiment 'p_2_10_hybrid_car_cost' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_10_hybrid_car_cost

Natural language statement: Write a method that helps a person decide whether to buy a hybrid car. Your methods's inputs should be: the cost of a new car, the estimated miles driven per year, the estimated gas price, the efficiency in miles per gallon, the estimated resale value after 5 years. Compute the total cost of owning the car for five years.

Method signature: p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real, mpg: real, resale_value: real) returns (total_cost: real)

Ensures

- `total_cost == new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value`

Requires

- `0.0 < mpg`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that helps a person decide whether to buy a hybrid car. Your methods's
→ inputs should be: the cost of a new car, the estimated miles driven per year, the
→ estimated gas price, the efficiency in miles per gallon, the estimated resale
→ value after 5 years. Compute the total cost of owning the car for five years.

The signature should be:

`p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real, mpg:
→ real, resale_value: real) returns (total_cost: real)`

The method should respect the following contract:

`requires 0.0 < mpg, ensures total_cost == new_car_cost + (miles_per_year / mpg) *
→ gas_price * 5.0 - resale_value`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdhuHQBShzgse0HJWxh7yFcHhL4SF

Model: gpt-5.5-2026-04-23

Created at: 1780484750

Finish reason: stop

Usage: 470 tokens in, and 43 tokens out

```
{  
  total_cost := new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value;  
}
```

Final Program

```
method p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real,  
→ mpg: real, resale_value: real) returns (total_cost: real)  
  requires 0.0 < mpg  
  ensures total_cost == new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 -  
→ resale_value  
{  
  total_cost := new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value;  
}
```

Total Token Usage

Input tokens: 470

Output tokens: 43

Reasoning tokens: 0

Sum of 'total tokens': 513

Experiment Timings

Overall Experiment started at 1780484749177, ended at 1780484756573, lasting 7396ms (7.40 seconds)

Iteration #1 started at 1780484749178, ended at 1780484756573, lasting 7395ms (7.40 seconds)

Experiment 'p_2_1_letter_dimensions' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 1

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_1_letter_dimensions

Natural language statement: Write a method that computes and returns the dimensions of a letter-size (8.5 x 11 inches) sheet of paper in millimeters. There are 25.4 millimeters per inch. Use constants and comments in your method.

Method signature: p_2_1_letter_dimensions() returns (width: real, height: real)

Ensures

- width == 8.5 * 25.4
- height == 11.0 * 25.4

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that computes and returns the dimensions of a letter-size (8.5 x 11 inches
→) sheet of paper in millimeters. There are 25.4 millimeters per inch. Use
→ constants and comments in your method.

The signature should be:

`p_2_1_letter_dimensions()` returns (width: real, height: real)

The method should respect the following contract:

`ensures width == 8.5 * 25.4, ensures height == 11.0 * 25.4`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdVr9kF8FvziabRhjacnNUi9l3dz

Model: gpt-5.5-2026-04-23

Created at: 1780484003

Finish reason: stop

Usage: 362 tokens in, and 289 tokens out

```
{
  // Conversion factor: one inch is exactly 25.4 millimeters.
  const millimetersPerInch: real := 25.4;

  // Letter-size paper dimensions in inches.
  const letterWidthInches: real := 8.5;
  const letterHeightInches: real := 11.0;

  // Convert the dimensions from inches to millimeters.
  width := letterWidthInches * millimetersPerInch;
  height := letterHeightInches * millimetersPerInch;
}
```

Program → GenAI

When using dafny resolve, the below error is emitted and resolve fails:

`problem_data/p_2_1_letter_dimensions/program.dfy(6,2): Error: rbrace expected`

```

|
6 | const millimetersPerInch: real := 25.4;
| ^

problem_data/p_2_1_letter_dimensions/program.dfy(6,40): Warning: deprecated style: a semi
  ↳ -colon is not needed here
|
6 | const millimetersPerInch: real := 25.4;
| ^

problem_data/p_2_1_letter_dimensions/program.dfy(9,38): Warning: deprecated style: a semi
  ↳ -colon is not needed here
|
9 | const letterWidthInches: real := 8.5;
| ^

problem_data/p_2_1_letter_dimensions/program.dfy(10,40): Warning: deprecated style: a
  ↳ semi-colon is not needed here
|
10 | const letterHeightInches: real := 11.0;
| ^

problem_data/p_2_1_letter_dimensions/program.dfy(13,2): Error: this symbol not expected
  ↳ in Dafny
|
13 | width := letterWidthInches * millimetersPerInch;
| ^

2 parse errors detected in program.dfy

Correct the error in code by altering only the method body. Produce and show only the
  ↳ Dafny body, including the curly braces that surround it. Do not show the signature
  ↳ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

```

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdVzVxuSjQMztUV3UTWUC6OOwCZh

Model: gpt-5.5-2026-04-23

Created at: 1780484011

Finish reason: stop

Usage: 1013 tokens in, and 562 tokens out

```

{
  // Conversion factor: one inch is exactly 25.4 millimeters.
  var millimetersPerInch: real := 25.4;

  // Letter-size paper dimensions in inches.
  var letterWidthInches: real := 8.5;
  var letterHeightInches: real := 11.0;

```

```
// Convert the dimensions from inches to millimeters.  
width := letterWidthInches * millimetersPerInch;  
height := letterHeightInches * millimetersPerInch;  
}
```

Final Program

```
method p_2_1_letter_dimensions() returns (width: real, height: real)  
  ensures width == 8.5 * 25.4  
  ensures height == 11.0 * 25.4  
{  
  // Conversion factor: one inch is exactly 25.4 millimeters.  
  var millimetersPerInch: real := 25.4;  
  
  // Letter-size paper dimensions in inches.  
  var letterWidthInches: real := 8.5;  
  var letterHeightInches: real := 11.0;  
  
  // Convert the dimensions from inches to millimeters.  
  width := letterWidthInches * millimetersPerInch;  
  height := letterHeightInches * millimetersPerInch;  
}
```

Total Token Usage

Input tokens: 1375

Output tokens: 851

Reasoning tokens: 577

Sum of 'total tokens': 2226

Experiment Timings

Overall Experiment started at 1780484002399, ended at 1780484026463, lasting 24064ms (24.06 seconds)

Iteration #1 started at 1780484002407, ended at 1780484011342, lasting 8935ms (8.94 seconds)

Iteration #2 started at 1780484011343, ended at 1780484026462, lasting 15119ms (15.12 seconds)

Experiment 'p_2_2_letter_perimeter' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_2_letter_perimeter

Natural language statement: Write a method that computes and returns the perimeter of a letter-size (8.5 x 11 inches) sheet of paper and the length of its diagonal with a residual error less than 0.01.

Method signature: p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)

Ensures

- `perimeter == (8.5 + 11.0) * 2.0`
- `-0.01 < (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) < 0.01`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that computes and returns the perimeter of a letter-size (8.5 x 11 inches)
→ sheet of paper and the length of its diagonal with a residual error less than
→ 0.01.

The signature should be:

`p_2_2_letter_perimeter()` returns (perimeter: real, diagonal: real)

The method should respect the following contract:

ensures perimeter == (8.5 + 11.0) * 2.0, ensures -0.01 < (diagonal * diagonal) - ((8.5 *
→ 8.5) + (11.0 * 11.0)) < 0.01

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdWFTIFnFWoGbq4TxzFMU97pzN8X

Model: gpt-5.5-2026-04-23

Created at: 1780484027

Finish reason: stop

Usage: 420 tokens in, and 1677 tokens out

```
{
  perimeter := (8.5 + 11.0) * 2.0;
  diagonal := 13.9014;

  assert diagonal * diagonal == 193.24892196;
  assert (8.5 * 8.5) + (11.0 * 11.0) == 193.25;
  assert (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) == -0.00107804;
  assert -0.01 < -0.00107804;
  assert -0.00107804 < 0.01;
}
```

Final Program

```
method p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)
  ensures perimeter == (8.5 + 11.0) * 2.0
  ensures -0.01 < (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) < 0.01
```

```
{
  perimeter := (8.5 + 11.0) * 2.0;
  diagonal := 13.9014;

  assert diagonal * diagonal == 193.24892196;
  assert (8.5 * 8.5) + (11.0 * 11.0) == 193.25;
  assert (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) == -0.00107804;
  assert -0.01 < -0.00107804;
  assert -0.00107804 < 0.01;
}
```

Total Token Usage

Input tokens: 420

Output tokens: 1677

Reasoning tokens: 1501

Sum of 'total tokens': 2097

Experiment Timings

Overall Experiment started at 1780484027517, ended at 1780484063445, lasting 35928ms (35.93 seconds)

Iteration #1 started at 1780484027518, ended at 1780484063444, lasting 35926ms (35.93 seconds)

Experiment ‘p_2_3_power_functions’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_3_power_functions

Natural language statement: Write a method that takes a number and returns its square, cube, and fourth power.

Method signature: p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)

Ensures

- `sq == x * x`
- `cube == x * x * x`
- `fourth == x * x * x * x`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a number and returns its square, cube, and fourth power.

The signature should be:

`p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)`

The method should respect the following contract:

`ensures sq == x * x, ensures cube == x * x * x, ensures fourth == x * x * x * x`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdWqLx2ApIMI5kLQ4WxYxgXpmKDr

Model: gpt-5.5-2026-04-23

Created at: 1780484064

Finish reason: stop

Usage: 318 tokens in, and 116 tokens out

```
{
  sq := x * x;
  cube := x * x * x;
  fourth := x * x * x * x;
}
```

Final Program

```
method p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)
  ensures sq == x * x
  ensures cube == x * x * x
  ensures fourth == x * x * x * x
{
  sq := x * x;
  cube := x * x * x;
  fourth := x * x * x * x;
}
```

Total Token Usage

Input tokens: 318

Output tokens: 116

Reasoning tokens: 66

Sum of 'total tokens': 434

Experiment Timings

Overall Experiment started at 1780484064334, ended at 1780484070843, lasting 6509ms (6.51 seconds)

Iteration #1 started at 1780484064336, ended at 1780484070843, lasting 6507ms (6.51 seconds)

Experiment 'p_2_4_a_sum' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_a_sum

Natural language statement: Write a method that takes two integers and then returns the sum.

Method signature: p_2_4_a_sum(x: int, y: int) returns (result: int)

Ensures

- `result == x + y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the sum.

The signature should be:

`p_2_4_a_sum(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x + y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdWxY2idkpPNsFIIl9RLeAllR2uU

Model: gpt-5.5-2026-04-23

Created at: 1780484071

Finish reason: stop

Usage: 260 tokens in, and 56 tokens out

```
{  
  result := x + y;  
}
```

Final Program

```
method p_2_4_a_sum(x: int, y: int) returns (result: int)  
  ensures result == x + y  
{  
  result := x + y;  
}
```

Total Token Usage

Input tokens: 260

Output tokens: 56

Reasoning tokens: 28

Sum of ‘total tokens’: 316

Experiment Timings

Overall Experiment started at 1780484071592, ended at 1780484077109, lasting 5517ms (5.52 seconds)

Iteration #1 started at 1780484071593, ended at 1780484077108, lasting 5515ms (5.52 seconds)

Experiment 'p_2_4_b_difference' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_b_difference

Natural language statement: Write a method that takes two integers and then returns the difference.

Method signature: p_2_4_b_difference(x: int, y: int) returns (result: int)

Ensures

- `result == x - y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the difference.

The signature should be:

`p_2_4_b_difference(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x - y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdX4Z5MruoGpGxhwByMyCxV2LGfI

Model: gpt-5.5-2026-04-23

Created at: 1780484078

Finish reason: stop

Usage: 260 tokens in, and 22 tokens out

```
{  
  result := x - y;  
}
```

Final Program

```
method p_2_4_b_difference(x: int, y: int) returns (result: int)  
  ensures result == x - y  
{  
  result := x - y;  
}
```

Total Token Usage

Input tokens: 260

Output tokens: 22

Reasoning tokens: 0

Sum of ‘total tokens’: 282

Experiment Timings

Overall Experiment started at 1780484077868, ended at 1780484083041, lasting 5173ms (5.17 seconds)

Iteration #1 started at 1780484077869, ended at 1780484083040, lasting 5171ms (5.17 seconds)

Experiment 'p_2_4_c_product' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_c_product

Natural language statement: Write a method that takes two integers and then returns the product.

Method signature: p_2_4_c_product(x: int, y: int) returns (result: int)

Ensures

- `result == x * y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the product.

The signature should be:

`p_2_4_c_product(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x * y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXArU2fyAE9SGZRTdYkcmF6PW11

Model: gpt-5.5-2026-04-23

Created at: 1780484084

Finish reason: stop

Usage: 260 tokens in, and 22 tokens out

```
{  
  result := x * y;  
}
```

Final Program

```
method p_2_4_c_product(x: int, y: int) returns (result: int)  
  ensures result == x * y  
{  
  result := x * y;  
}
```

Total Token Usage

Input tokens: 260

Output tokens: 22

Reasoning tokens: 0

Sum of ‘total tokens’: 282

Experiment Timings

Overall Experiment started at 1780484083808, ended at 1780484089187, lasting 5379ms (5.38 seconds)

Iteration #1 started at 1780484083810, ended at 1780484089187, lasting 5377ms (5.38 seconds)

Experiment 'p_2_4_d_avarage' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_d_avarage

Natural language statement: Write a method that takes two integers and then returns the average.

Method signature: p_2_4_d_avarage(x: int, y: int) returns (result: real)

Ensures

- `result == (x + y) as real / 2.0`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the average.

The signature should be:

`p_2_4_d_avarage(x: int, y: int) returns (result: real)`

The method should respect the following contract:

`ensures result == (x + y) as real / 2.0`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXGWtc5XMIPipHaBzpVQIFNAYFI

Model: gpt-5.5-2026-04-23

Created at: 1780484090

Finish reason: stop

Usage: 282 tokens in, and 269 tokens out

```
{  
  result := ((x + y) as real) / 2.0;  
}
```

Final Program

```
method p_2_4_d_avarage(x: int, y: int) returns (result: real)  
  ensures result == (x + y) as real / 2.0  
{  
  result := ((x + y) as real) / 2.0;  
}
```

Total Token Usage

Input tokens: 282

Output tokens: 269

Reasoning tokens: 231

Sum of ‘total tokens’: 551

Experiment Timings

Overall Experiment started at 1780484089885, ended at 1780484099892, lasting 10007ms (10.01 seconds)

Iteration #1 started at 1780484089885, ended at 1780484099892, lasting 10007ms (10.01 seconds)

Experiment 'p_2_4_e_distance' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_e_distance

Natural language statement: Write a method that takes two integers and then returns the distance (absolute value of the difference).

Method signature: p_2_4_e_distance(x: int, y: int) returns (result: int)

Ensures

- `result >= 0`
- `if x >= y then result == x - y else result == y - x`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the distance (absolute value of
→ the difference).

The signature should be:

`p_2_4_e_distance(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result >= 0, ensures if x >= y then result == x - y else result == y - x`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXRVC41mAHgIbp6T33DOp1e3xaC

Model: gpt-5.5-2026-04-23

Created at: 1780484101

Finish reason: stop

Usage: 306 tokens in, and 74 tokens out

```
{
  if x >= y {
    result := x - y;
  } else {
    result := y - x;
  }
}
```

Final Program

```
method p_2_4_e_distance(x: int, y: int) returns (result: int)
  ensures result >= 0
  ensures if x >= y then result == x - y else result == y - x
{
  if x >= y {
    result := x - y;
  } else {
    result := y - x;
  }
}
```

```
}  
}
```

Total Token Usage

Input tokens: 306

Output tokens: 74

Reasoning tokens: 23

Sum of 'total tokens': 380

Experiment Timings

Overall Experiment started at 1780484100565, ended at 1780484108038, lasting 7473ms (7.47 seconds)

Iteration #1 started at 1780484100565, ended at 1780484108033, lasting 7468ms (7.47 seconds)

Experiment 'p_2_4_f_max' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_f_max

Natural language statement: Write a method that takes two integers and then returns the maximum (the larger of the two).

Method signature: p_2_4_f_max(x: int, y: int) returns (result: int)

Ensures

- `result == x || result == y`
- `if x >= y then result == x else result == y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the maximum (the larger of the
→ two).

The signature should be:

`p_2_4_f_max(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x || result == y, ensures if x >= y then result == x else result == y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXZU22pP0LubprgaiqaNQJESL3r

Model: gpt-5.5-2026-04-23

Created at: 1780484109

Finish reason: stop

Usage: 304 tokens in, and 68 tokens out

```
{
  if x >= y {
    result := x;
  } else {
    result := y;
  }
}
```

Final Program

```
method p_2_4_f_max(x: int, y: int) returns (result: int)
  ensures result == x || result == y
  ensures if x >= y then result == x else result == y
{
  if x >= y {
    result := x;
  } else {
    result := y;
  }
}
```

```
}  
}
```

Total Token Usage

Input tokens: 304

Output tokens: 68

Reasoning tokens: 21

Sum of 'total tokens': 372

Experiment Timings

Overall Experiment started at 1780484108752, ended at 1780484115065, lasting 6313ms (6.31 seconds)

Iteration #1 started at 1780484108756, ended at 1780484115065, lasting 6309ms (6.31 seconds)

Experiment 'p_2_4_g_min' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_g_min

Natural language statement: Write a method that takes two integers and then returns the minimum (the smaller of the two).

Method signature: p_2_4_g_min(x: int, y: int) returns (result: int)

Ensures

- `result == x || result == y`
- `if x <= y then result == x else result == y`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the minimum (the smaller of the
→ two).

The signature should be:

`p_2_4_g_min(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result == x || result == y, ensures if x <= y then result == x else result == y`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXgrBtIgMPUYrkGlpEYIik5MJ7M

Model: gpt-5.5-2026-04-23

Created at: 1780484116

Finish reason: stop

Usage: 304 tokens in, and 75 tokens out

```
{
  if x <= y {
    result := x;
  } else {
    result := y;
  }
}
```

Final Program

```
method p_2_4_g_min(x: int, y: int) returns (result: int)
  ensures result == x || result == y
  ensures if x <= y then result == x else result == y
{
  if x <= y {
    result := x;
  } else {
    result := y;
  }
}
```

```
}  
}
```

Total Token Usage

Input tokens: 304

Output tokens: 75

Reasoning tokens: 28

Sum of 'total tokens': 379

Experiment Timings

Overall Experiment started at 1780484115716, ended at 1780484122517, lasting 6801ms (6.80 seconds)

Iteration #1 started at 1780484115716, ended at 1780484122517, lasting 6801ms (6.80 seconds)

Experiment ‘p_2_6_convert’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_6_convert

Natural language statement: Write a method that takes a measurement in meters and then converts it to miles, feet, and inches.

Method signature: p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)

Ensures

- `x / 0.000621371 == miles`
- `x / 3.28084 == feet`
- `x / 39.3701 == inches`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a measurement in meters and then converts it to miles, feet,
↪ and inches.

The signature should be:

`p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)`

The method should respect the following contract:

`ensures x / 0.000621371 == miles, ensures x / 3.28084 == feet, ensures x / 39.3701 ==`
↪ `inches`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXnMCqOMjI937jJwgEctxeUfHZL

Model: gpt-5.5-2026-04-23

Created at: 1780484123

Finish reason: stop

Usage: 338 tokens in, and 190 tokens out

```
{  
  miles := x / 0.000621371;  
  feet := x / 3.28084;  
  inches := x / 39.3701;  
}
```

Final Program

```
method p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)  
  ensures x / 0.000621371 == miles  
  ensures x / 3.28084 == feet  
  ensures x / 39.3701 == inches  
{  
  miles := x / 0.000621371;  
  feet := x / 3.28084;  
  inches := x / 39.3701;  
}
```

Total Token Usage

Input tokens: 338

Output tokens: 190

Reasoning tokens: 133

Sum of 'total tokens': 528

Experiment Timings

Overall Experiment started at 1780484123148, ended at 1780484131028, lasting 7880ms (7.88 seconds)

Iteration #1 started at 1780484123148, ended at 1780484131028, lasting 7880ms (7.88 seconds)

Experiment 'p_2_7_circle_properties' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_7_circle_properties

Natural language statement: Write a method that takes a radius and then returns the area and circumference of a circle with that radius and the volume and surface area of a sphere with that radius.

Method signature: p_2_7_circle_properties(radius: real) returns (circumference: real, area: real, volume: real, surface_area: real)

Ensures

- `circumference == 2.0 * 3.14159 * radius`
- `area == 3.14159 * radius * radius`
- `volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius`
- `surface_area == 4.0 * 3.14159 * radius * radius`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a radius and then returns the area and circumference of a
→ circle with that radius and the volume and surface area of a sphere with that
→ radius.

The signature should be:

p_2_7_circle_properties(radius: real) returns (circumference: real, area: real, volume:
→ real, surface_area: real)

The method should respect the following contract:

ensures circumference == 2.0 * 3.14159 * radius, ensures area == 3.14159 * radius *
→ radius, ensures volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius,
→ ensures surface_area == 4.0 * 3.14159 * radius * radius

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXvR8NHgstmZSiH7DF20dhpOxkU

Model: gpt-5.5-2026-04-23

Created at: 1780484131

Finish reason: stop

Usage: 456 tokens in, and 324 tokens out

```
{  
  circumference := 2.0 * 3.14159 * radius;  
  area := 3.14159 * radius * radius;  
  volume := (4.0 / 3.0) * 3.14159 * radius * radius * radius;  
  surface_area := 4.0 * 3.14159 * radius * radius;  
}
```

Final Program

```
method p_2_7_circle_properties(radius: real) returns (circumference: real, area: real,  
→ volume: real, surface_area: real)  
  ensures circumference == 2.0 * 3.14159 * radius  
  ensures area == 3.14159 * radius * radius  
  ensures volume == (4.0 / 3.0) * 3.14159 * radius * radius * radius
```

```
        ensures surface_area == 4.0 * 3.14159 * radius * radius
{
    circumference := 2.0 * 3.14159 * radius;
    area := 3.14159 * radius * radius;
    volume := (4.0 / 3.0) * 3.14159 * radius * radius * radius;
    surface_area := 4.0 * 3.14159 * radius * radius;
}
```

Total Token Usage

Input tokens: 456

Output tokens: 324

Reasoning tokens: 226

Sum of 'total tokens': 780

Experiment Timings

Overall Experiment started at 1780484131654, ended at 1780484142138, lasting 10484ms (10.48 seconds)

Iteration #1 started at 1780484131654, ended at 1780484142138, lasting 10484ms (10.48 seconds)

Experiment 'p_2_8_rectangle_properties' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 5

Hard fails (resolution): 1

Soft fails (resolution): 1

Verification attempts: 3

Problem Specification

Problem name: p_2_8_rectangle_properties

Natural language statement: Write a method that takes the lengths of the sides of a rectangle. Then return the area and perimeter of the rectangle and the length of the diagonal (use the Pythagorean theorem) with a residual error less than 0.01.

Method signature: p_2_8_rectangle_properties(length: real, width: real) returns (area: real, perimeter: real, diagonal: real)

Ensures

- `area == length * width`
- `perimeter == 2.0 * (length + width)`
- `-0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) < 0.01`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes the lengths of the sides of a rectangle. Then return the area
→ and perimeter of the rectangle and the length of the diagonal (use the Pythagorean
→ theorem) with a residual error less than 0.01.

The signature should be:

p_2_8_rectangle_properties(length: real, width: real) returns (area: real, perimeter:
→ real, diagonal: real)

The method should respect the following contract:

ensures area == length * width, ensures perimeter == 2.0 * (length + width), ensures
→ -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) < 0.01

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdY7yGrU6LXn0OV8GZZlnC3Mbbbx

Model: gpt-5.5-2026-04-23

Created at: 1780484143

Finish reason: stop

Usage: 434 tokens in, and 1162 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := length * length + width * width;
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;
  assert exists d: real :: d * d == s;
  diagonal :| diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
}
```

Program → GenAI

When using dafny resolve, the below error is emitted and resolve fails:

```
problem_data/p_2_8_rectangle_properties/program.dfy(13,9): Warning: Could not find a
  ↳ trigger for this quantifier. Without a trigger, the quantifier may cause brittle
  ↳ verification. To silence this warning, add an explicit trigger using the {:trigger
  ↳ } attribute. For more information, see the section on quantifier instantiation
  ↳ rules in the reference manual.
```

```
|
13 | assert exists d: real :: d * d == s;
| ~~~~~
```

Compilation failed because warnings were found and --allow-warnings is false

Correct the error in code by altering only the method body. Produce and show only the

- ↳ Dafny body, including the curly braces that surround it. Do not show the signature
- ↳ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdYaWXIAScgiigIGMQgYxXyecUg5

Model: gpt-5.5-2026-04-23

Created at: 1780484172

Finish reason: stop

Usage: 730 tokens in, and 3069 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := length * length + width * width;
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;
  diagonal :| diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_2_8_rectangle_properties/program.dfy(13,11): Error: cannot establish the
  ↳ existence of LHS values that satisfy the such-that predicate. Note, no trigger was
  ↳ found for the such-that predicate, which may be the reason the proof failed. To
  ↳ give a trigger explicitly, use the {:trigger} attribute. For more information, see
  ↳ the section on quantifier instantiation rules in the reference manual.
|
13 | diagonal :| diagonal * diagonal == s;
| ^^
```


Dafny program verifier finished with 0 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
→ Dafny body, including the curly braces that surround it. Do not show the signature
→ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdZjWgYiKUif4jtzuy5SssTEQlrD

Model: gpt-5.5-2026-04-23

Created at: 1780484243

Finish reason: stop

Usage: 1046 tokens in, and 5312 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := length * length + width * width;
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;
  assert exists d: real {:trigger d * d} :: d * d == s;
  diagonal :| diagonal * diagonal == s;
  assert (diagonal * diagonal) - s == 0.0;
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_2_8_rectangle_properties/program.dfy(13,2): Error: assertion might not
  → hold
  |
13 | assert exists d: real {:trigger d * d} :: d * d == s;
  | ~~~~~
```

Dafny program verifier finished with 0 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
→ Dafny body, including the curly braces that surround it. Do not show the signature
→ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dmdbk8uEROJ2HPdelHnH2nfajnp5

Model: gpt-5.5-2026-04-23

Created at: 1780484368

Finish reason: stop

Usage: 1272 tokens in, and 4269 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := length * length + width * width;
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;

  diagonal := Std.Math.Sqrt(s);
  assert diagonal * diagonal == s;
  assert -0.01 < (diagonal * diagonal) - s < 0.01;
}
```

Program → GenAI

When using dafny resolve, the below error is emitted and resolve fails:

problem_data/p_2_8_rectangle_properties/program.dfy(14,14): Error: unresolved identifier:

```
  ↪ Std
  |
14 | diagonal := Std.Math.Sqrt(s);
  | ~~~
```

1 resolution/type errors detected in program.dfy

Correct the error in code by altering only the method body. Produce and show only the

- ↪ Dafny body, including the curly braces that surround it. Do not show the signature
- ↪ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmddS0mYKam1Tr9pXBKZQg1IoVdKW

Model: gpt-5.5-2026-04-23

Created at: 1780484474

Finish reason: stop

Usage: 1506 tokens in, and 11054 tokens out

```
{
  area := length * width;
  perimeter := 2.0 * (length + width);

  var s := length * length + width * width;
  assert 0.0 <= length * length;
  assert 0.0 <= width * width;
  assert 0.0 <= s;
```

```

var hi := 1.0;
while hi * hi <= s
  invariant 1.0 <= hi
  invariant 0.0 <= s
  decreases s - hi
{
  assert hi <= hi * hi;
  assert hi <= s;
  hi := 2.0 * hi;
}

var lo := 0.0;
var H := hi;
var factor := 1.0;
var target := 256.0 * H * H;

while factor < target
  invariant 1.0 <= H
  invariant 1.0 <= factor
  invariant 0.0 <= lo <= hi <= H
  invariant lo * lo <= s < hi * hi
  invariant (hi - lo) * factor <= H
  invariant target == 256.0 * H * H
  decreases target - factor
{
  var oldLo := lo;
  var oldHi := hi;
  var oldFactor := factor;
  var mid := (oldLo + oldHi) / 2.0;

  assert oldLo <= mid <= oldHi;
  if mid * mid <= s {
    lo := mid;
  } else {
    hi := mid;
  }

  assert hi - lo == (oldHi - oldLo) / 2.0;
  factor := 2.0 * oldFactor;
  assert (hi - lo) * factor == (oldHi - oldLo) * oldFactor;
}

assert target <= factor;
assert 0.0 <= hi - lo;
assert (hi - lo) * target <= (hi - lo) * factor;
assert (hi - lo) * target <= H;
assert 0.0 < H;
assert target == 256.0 * H * H;
assert ((hi - lo) * 256.0 * H) * H == (hi - lo) * target;
assert (hi - lo) * 256.0 * H <= 1.0;

```

```

assert hi + lo <= 2.0 * H;
assert (hi - lo) * (hi + lo) <= (hi - lo) * (2.0 * H);
assert (hi - lo) * (hi + lo) <= 1.0 / 128.0;
assert 1.0 / 128.0 < 0.01;
assert hi * hi - lo * lo == (hi - lo) * (hi + lo);
assert hi * hi - lo * lo < 0.01;

diagonal := lo;
assert diagonal * diagonal <= s;
assert s < hi * hi;
assert s - diagonal * diagonal < 0.01;
assert -0.01 < diagonal * diagonal - s;
assert diagonal * diagonal - s < 0.01;
}

```

Final Program

```

method p_2_8_rectangle_properties(length: real, width: real) returns (area: real,
    ↪ perimeter: real, diagonal: real)
    ensures area == length * width
    ensures perimeter == 2.0 * (length + width)
    ensures -0.01 < (diagonal * diagonal) - ((length * length) + (width * width)) <
        ↪ 0.01
{
    area := length * width;
    perimeter := 2.0 * (length + width);

    var s := length * length + width * width;
    assert 0.0 <= length * length;
    assert 0.0 <= width * width;
    assert 0.0 <= s;

    var hi := 1.0;
    while hi * hi <= s
        invariant 1.0 <= hi
        invariant 0.0 <= s
        decreases s - hi
    {
        assert hi <= hi * hi;
        assert hi <= s;
        hi := 2.0 * hi;
    }

    var lo := 0.0;
    var H := hi;
    var factor := 1.0;
    var target := 256.0 * H * H;

    while factor < target
        invariant 1.0 <= H

```

```

invariant 1.0 <= factor
invariant 0.0 <= lo <= hi <= H
invariant lo * lo <= s < hi * hi
invariant (hi - lo) * factor <= H
invariant target == 256.0 * H * H
decreases target - factor
{
  var oldLo := lo;
  var oldHi := hi;
  var oldFactor := factor;
  var mid := (oldLo + oldHi) / 2.0;

  assert oldLo <= mid <= oldHi;
  if mid * mid <= s {
    lo := mid;
  } else {
    hi := mid;
  }

  assert hi - lo == (oldHi - oldLo) / 2.0;
  factor := 2.0 * oldFactor;
  assert (hi - lo) * factor == (oldHi - oldLo) * oldFactor;
}

assert target <= factor;
assert 0.0 <= hi - lo;
assert (hi - lo) * target <= (hi - lo) * factor;
assert (hi - lo) * target <= H;
assert 0.0 < H;
assert target == 256.0 * H * H;
assert ((hi - lo) * 256.0 * H) * H == (hi - lo) * target;
assert (hi - lo) * 256.0 * H <= 1.0;
assert hi + lo <= 2.0 * H;
assert (hi - lo) * (hi + lo) <= (hi - lo) * (2.0 * H);
assert (hi - lo) * (hi + lo) <= 1.0 / 128.0;
assert 1.0 / 128.0 < 0.01;
assert hi * hi - lo * lo == (hi - lo) * (hi + lo);
assert hi * hi - lo * lo < 0.01;

diagonal := lo;
assert diagonal * diagonal <= s;
assert s < hi * hi;
assert s - diagonal * diagonal < 0.01;
assert -0.01 < diagonal * diagonal - s;
assert diagonal * diagonal - s < 0.01;
}

```

Total Token Usage

Input tokens: 4988

Output tokens: 24866

Reasoning tokens: 23352

Sum of ‘total tokens’: 29854

Experiment Timings

Overall Experiment started at 1780484142817, ended at 1780484748023, lasting 605206ms (605.21 seconds)

Iteration #4 started at 1780484367476, ended at 1780484473664, lasting 106188ms (106.19 seconds)

Iteration #5 started at 1780484473664, ended at 1780484748023, lasting 274359ms (274.36 seconds)

Iteration #1 started at 1780484142817, ended at 1780484171937, lasting 29120ms (29.12 seconds)

Iteration #2 started at 1780484171938, ended at 1780484243481, lasting 71543ms (71.54 seconds)

Iteration #3 started at 1780484243482, ended at 1780484367476, lasting 123994ms (123.99 seconds)

